P P SAVANI UNIVERSITY

SCHOOL OF ENGINEERING

B.TECH FINAL PROJECT REPORT

(Hard-Bound Project Report Format)

PARKSMART: REAL-TIME SMART PARKING RESERVATION & AUTOMATED SECURITY GATE ACCESS MANAGEMENT SYSTEM

Particular	Details
Project Title	ParkSmart: Real-Time Smart Parking Reservation & Automated Security Gate Access Management System
Student Name	[Student Name]
Enrollment Number	[Enrollment Number]
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Department / Domain	Full-Stack Web Development & Smart Transportation Systems
Mentor (Company)	Not Applicable / Industry Guide
Mentor (Institute)	[Internal Guide / Faculty Mentor Name]
Academic Year	2025–2026

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CERTIFICATE OF COMPLETION

This is to certify that **[STUDENT NAME]**, Enrollment No.: **[ENROLLMENT NUMBER]**, student of **B.Tech Computer Engineering, 8th Semester**, has successfully completed the Final Year Project titled:

"PARKSMART: REAL-TIME SMART PARKING RESERVATION & AUTOMATED SECURITY GATE ACCESS MANAGEMENT SYSTEM"

during the academic term at **P P Savani University** under the supervision and technical guidance of **[Institute Mentor Name]**.

The project work, architectural implementation, security hardening, testing and validation, system documentation, and practical demonstrations were carried out satisfactorily as per the university engineering requirements.

Date of Issue: [DD / MM / YYYY]

_____ _____

Mentor (Company / External) **Mentor (Institute / Internal)**

DECLARATION

I hereby declare that this project report entitled **"ParkSmart: Real-Time Smart Parking Reservation & Automated Security Gate Access Management System"** is an authentic record of the genuine project work carried out by me as part of the curriculum for the award of the degree of **Bachelor of Technology in Computer Engineering** at **School of Engineering, P P Savani University**.

I confirm that this work has not been submitted previously to any other university or institution for the award of any degree or diploma. All external references, literature sources, and third-party software libraries have been duly acknowledged and cited in accordance with standard academic practices.

Signature of the Student: _____

Name: [Student Name]

Enrollment Number: [Enrollment Number]

Date: [DD / MM / YYYY]

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I express my deepest gratitude to **P P Savani University** and the **School of Engineering** for providing state-of-the-art laboratory infrastructure, computational facilities, and an encouraging learning atmosphere to complete this project.

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ABSTRACT / EXECUTIVE SUMMARY

Rapid urbanization and rising vehicle ownership have resulted in severe parking congestion, time loss, and air pollution in commercial zones, universities, and malls. Traditional parking management systems suffer from serious bottlenecks: reliance on paper tickets, lack of real-time slot visibility, manual fee collection, and vulnerable static passes that can be duplicated or shared.

ParkSmart is an end-to-end, responsive smart parking reservation and security gate management platform developed on the **MERN stack (MongoDB, Express.js, React with TypeScript, Node.js)**. The system provides three integrated web portals tailored for Drivers, Parking Administrators, and Security Gate Staff.

Key engineering contributions of ParkSmart include:

- **Interactive 2D Slot Matrix:** Provides drivers with real-time visual floor plans categorizing bays by vehicle type (Two-Wheeler, Four-Wheeler, EV Charging) with live occupancy status.
- **Cryptographic Dynamic QR Passes:** Employs a time-synchronized HMAC-SHA256 token rotation algorithm (refreshing every 30 seconds) with a ±15-minute drift tolerance window to prevent screenshot re-entry fraud.
- **Smart Auto Gate Security Desk:** Features an in-browser scanner with multi-camera fallback (rear/webcam/USB), screenshot photo upload decoding, continuous scanning mode, and automatic entry/exit state detection.
- **Automated Overstay Billing:** Accurately calculates parking durations, base rates, and penalty fees for overdue vehicles without manual intervention.

Performance testing demonstrated an average QR pass verification speed of under 1.5 seconds, zero TypeScript compilation errors, and complete cross-platform responsiveness across mobile and desktop devices.

Keywords: Smart Parking System, MERN Stack, Dynamic QR Pass, HMAC-SHA256 Verification, Real-Time Slot Reservation, Smart Auto Gate Desk, Overstay Billing.

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LIST OF ABBREVIATIONS / ACRONYMS

Abbreviation	Full Form / Meaning
MERN	MongoDB, Express.js, React.js, Node.js
QR Code	Quick Response Code
HMAC	Hash-Based Message Authentication Code
SHA-256	Secure Hash Algorithm 256-Bit
JWT	JSON Web Token
REST	Representational State Transfer
API	Application Programming Interface
UI / UX	User Interface / User Experience
RBAC	Role-Based Access Control
SPA	Single Page Application
ANPR	Automatic Number Plate Recognition
CORS	Cross-Origin Resource Sharing
NoSQL	Non-Relational Structured Query Language
DOM	Document Object Model

CHAPTER 1: INTRODUCTION

1.1 Background of the Project

Urban traffic congestion has become a critical challenge worldwide. Studies indicate that up to 30% of downtown traffic is caused solely by drivers searching for vacant parking spaces. Traditional parking systems rely heavily on manual attendants, printed thermal paper tickets, and manual cash handovers. This conventional setup results in long queues at entry points, billing inaccuracies, ticket loss, and no visibility into space availability prior to arrival.

1.2 Problem Statement

"To design, develop, and evaluate a responsive full-stack smart parking reservation and security gate access system that provides real-time slot booking for motorists, cryptographically secure dynamic QR gate access for security staff, and comprehensive financial/operational analytics for parking administrators."

1.3 Need / Motivation

- **Time and Fuel Efficiency:** Eliminates blind cruising for parking by letting motorists reserve spaces in advance.
- **Gate Throughput:** Fast, in-browser QR scanning reduces vehicle entry and exit time to under 2 seconds.
- **Fraud Prevention:** Standard static QR codes can be screenshotted and shared. A rotating dynamic HMAC token ensures single-use validity.
- **Automated Revenue Control:** Replaces manual estimation with exact timestamped billing and automatic overstay fee calculation.

1.4 Objectives

1. Develop a modern, responsive single-page web app for slot discovery and booking.
2. Create an interactive visual floor map reflecting real-time bay occupancy.
3. Implement an HMAC-SHA256 rotating dynamic QR pass refreshed every 30 seconds.
4. Build a versatile Security Gate Desk with multi-camera fallback, continuous scanning, and Smart Auto Entry/Exit handling.
5. Design an Admin Dashboard for revenue monitoring, slot maintenance, and access auditing.

1.5 Scope Matrix

In Scope	Out of Scope	Constraints
• Driver registration & JWT login • Interactive visual floor grid • Dynamic rotating QR pass • In-browser camera gate scanner • Smart Auto Gate entry/exit • Overstay penalty calculations • Admin analytics & slot management	• Physical barrier motor wiring • Ultrasonic sensors per bay • Native banking payment sandbox (digital wallet simulated)	• Requires camera permission on client • Active internet connection for live cloud sync • Modern browser with WebRTC support

1.6 Expected Outcomes vs Final Outcomes

Feature	Planned Expected Outcome	Final Achieved Outcome
Slot Selection	Dropdown list of slot numbers	Visual 2D interactive grid with vehicle filters (2W, 4W, EV)
Digital Pass	Static QR with raw booking ID	Rotating Dynamic QR with HMAC-SHA256 signature
Gate Scanner	Basic webcam reader	Smart Auto mode, multi-camera selector, and photo upload
Overstay	Manual guard estimation	Automated duration calculation with hourly penalties

CHAPTER 2: PROJECT PROFILE & WORK ENVIRONMENT

2.1 Domain Overview

ParkSmart is situated in the domain of **Smart Transportation Systems & Full-Stack Web Engineering**. It bridges consumer-facing web reservation platforms with on-premise security operational desks.

2.2 Work Environment & Assigned Roles

- **Frontend Architecture:** Developed using React 18, TypeScript, and clean Vanilla CSS without bulky third-party UI dependencies.
- **Backend Services:** Built using Node.js and Express.js adhering to REST API principles with secure JWT middleware.
- **Database Engineering:** Designed with MongoDB Mongoose schemas supporting relational references and indexing.

CHAPTER 3: LITERATURE REVIEW & BACKGROUND STUDY

3.1 Literature Review

Existing parking systems fall into three categories: manual paper ticketing, sensor-intensive hardware parking, and static QR mobile apps. While sensor systems offer automation, their per-slot hardware maintenance makes them cost-prohibitive. Static QR systems are affordable but vulnerable to ticket sharing.

3.2 Comparative Analysis

Parameter	Manual Attendant	Sensor-Based IoT	ParkSmart Platform
Infrastructure Cost	Low initial, high labor	Extremely high per-bay sensors	Low (Pure Web Software)
Pre-Booking Facility	No	Limited	Yes (Live Visual Map)
Pass Security	Paper slips (easily lost)	RFID cards (clonable)	HMAC Dynamic QR (Anti-Fraud)

Gate Verification Speed	30–60 seconds	5–10 seconds	< 1.5 seconds

CHAPTER 4: REQUIREMENTS ANALYSIS

4.1 Functional Requirements

Req ID	Requirement Description	Priority	Status
FR-01	User registration, login, and JWT session authorization	High	Completed
FR-02	Real-time visual 2D parking bay matrix	High	Completed
FR-03	Slot reservation with vehicle type & duration filters	High	Completed
FR-04	HMAC-SHA256 Dynamic QR token generation (30s rotation)	High	Completed
FR-05	Security Desk camera scanner with Smart Auto Mode	High	Completed
FR-06	1-Click Gate mode recovery for active/confirmed vehicles	Medium	Completed
FR-07	Automated overstay penalty billing at checkout	High	Completed
FR-08	Admin dashboard for slot management and revenue logs	High	Completed

4.2 Non-Functional Requirements

Req ID	Category	Target Specification	Status
NFR-01	Performance	API response time < 200ms under standard loads	Met
NFR-02	Security	bcrypt password hashing (10 salt rounds), JWT expiration	Met
NFR-03	Responsiveness	Fluid layout across 320px to 1920px viewports	Met
NFR-04	Reliability	±15-minute clock drift tolerance for dynamic tokens	Met

4.3 Software & Hardware Stack

- **Development Stack:** Node.js v20+, React 18, TypeScript, Express.js, MongoDB Atlas.
- **Libraries:** `bcryptjs`, `jsonwebtoken`, `html5-qrcode`, `qrcode.react`.
- **Client Environment:** Chrome, Edge, Firefox, or Safari on Windows, Android, macOS, or iOS.

CHAPTER 5: METHODOLOGY & PLANNING

5.1 Agile Methodology

The project followed an iterative Agile Scrum methodology across four distinct development sprints:

- **Sprint 1:** System architecture, Mongoose schemas, and JWT authentication pipeline.
- **Sprint 2:** Visual slot grid interface, reservation controller, and booking state logic.
- **Sprint 3:** HMAC cryptographic engine, Dynamic QR generation, and in-browser camera scanning.
- **Sprint 4:** Smart Auto Gate mode, multi-camera fallback, continuous scanning debounce, and final testing.

CHAPTER 6: SYSTEM ANALYSIS & DESIGN

6.1 3-Tier Architecture

> **1. Presentation Tier (Frontend):** React 18 SPA with TypeScript, HTML5 camera streams, and responsive UI components.
> **2. Logic Tier (Backend):** Node.js / Express.js REST APIs with cryptographic validation and role authorization.
> **3. Data Tier (Database):** MongoDB Atlas storing Users, Slots, Bookings, SecurityLogs, and Blacklists.

6.2 Major Database Entities

Entity	Key Attributes	Purpose
User	`_id, name, email, password, role, vehicleNumber`	User credentials & role authorization
Slot	`_id, slotNumber, floor, type, status, hourlyRate`	Individual parking bay attributes
Booking	`_id, user, slot, startTime, endTime, status, qrToken`	Core reservation and check-in records

SecurityLog	`_id, bookingId, gateMode, timestamp, scannedBy`	Gate compliance & audit logging

CHAPTER 7: TECHNOLOGY STACK

Technology	Version	Purpose & Usage in Project
TypeScript / JS	ES2022	Strict static typing on frontend and dynamic backend handling
React	18.2	Dynamic single-page application component hierarchy
Node.js	20.x	High-concurrency asynchronous runtime for API endpoints
Express.js	4.18	RESTful routing, error handling, and security middleware
MongoDB Atlas	6.0+	Cloud NoSQL database with indexing and aggregation
Html5Qrcode	2.3+	Hardware webcam/rear camera video feed decoding
Node Crypto	Built-in	HMAC-SHA256 cryptographic dynamic signature generation

CHAPTER 8: SYSTEM IMPLEMENTATION

8.1 Dynamic QR Cryptographic Algorithm

To prevent pass duplication and unauthorized entry via screenshots, passes use a time-rotating token:

> **Token Format:** `PS-DYN|<bookingId>|<timeStep>|<hmacSignature>`
> • `timeStep = Math.floor(currentTime / 30000)` (Rotates every 30 seconds)
> • `hmacSignature = HMAC_SHA256(bookingId + ':' + timeStep, SECRET_KEY)`
> • Backend checks current timeStep ± 30 steps (±15 minutes) to handle client clock skew seamlessly.

8.2 Smart Auto Gate Security Desk

- **Direction Auto-Detection:** When scanning a pass, the server checks vehicle status:
  - If `status == 'confirmed'` ➜ Grants Entry, marks slot `occupied`, and stamps check-in time.

    - If `status == 'active'` ➜ Grants Exit, frees the slot, and calculates overstay fees if due.

- **1-Click Recovery:** If an active car is scanned in manual Entry mode, the UI displays driver details with a single **"Switch to EXIT & Allow Exit"** button.
- **Continuous Scan Mode:** Keeps the camera open for high-traffic gates with a 2.5-second debounce.

CHAPTER 9: TESTING, DEFECT RESOLUTION & VALIDATION

9.1 Functional & Security Test Cases

Test ID	Module	Scenario	Result	Status
TC-01	Auth	Login with incorrect password	401 Error	Passed
TC-02	Booking	Reserve already occupied slot	Disabled	Passed
TC-03	Dynamic QR	Validate token with clock drift	Verified	Passed
TC-04	Camera	Start on laptop without rear lens	Webcam fallback	Passed
TC-05	Smart Gate	Scan arriving and departing vehicle	Auto-routed	Passed

9.2 Major Defect Resolution Record

Issue ID	Reported Bug	Root Cause	Resolution
BUG-01	Camera closed after 2s showing "Already Inside"	Scanner stopped on reading active pass in entry mode	Implemented Smart Auto Gate Mode and debounce
BUG-02	Dynamic QR token hash mismatch	Frontend uppercase query validator corrupted hex hash	Implemented case-insensitive HMAC check
BUG-03	Camera startup error on laptops	Hardcoded rear camera constraint	Multi-level camera fallback hierarchy

CHAPTER 10: RESULTS & PERFORMANCE ANALYSIS

- **Verification Speed:** Average scan-to-gate decision latency: **1.2 – 1.5 seconds**.
- **Build Health:** 100% clean production build (0 TypeScript compilation errors).
- **Device Compatibility:** Successfully validated on Android Chrome, iOS Safari, and Desktop Chrome/Edge.

CHAPTER 11: CHALLENGES & IMPROVEMENTS

Key engineering challenges included handling diverse browser camera constraints, mobile-server clock discrepancies, and rapid continuous scanning at busy parking gates. All challenges were resolved through adaptive fallbacks and cryptographic drift tolerance.

CHAPTER 12: INDIVIDUAL CONTRIBUTION & LEARNINGS

- Designed complete frontend architecture using React 18, TypeScript, and Vanilla CSS.
- Created secure Node.js/Express backend APIs, dynamic HMAC generation, and MongoDB schemas.
- Acquired extensive practical experience in full-stack architecture, WebRTC camera APIs, and cryptographic security.

CHAPTER 13: LIMITATIONS & FUTURE SCOPE

- **Current Limitations:** Requires internet connection for real-time cloud database synchronization.
- **Future Enhancements:** Integration with physical ESP32 barrier servos, automatic AI license plate recognition (ANPR), and direct payment gateway integration (UPI / Razorpay).

CHAPTER 14: CONCLUSION

ParkSmart successfully replaces outdated manual parking ticketing with an automated, secure, and user-friendly web solution. The implementation achieved all functional goals including live slot visibility, dynamic fraud-proof passes, and instant gate verification.

CHAPTER 15: REFERENCES (APA 7TH EDITION)

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FINAL HARD-BOUND SUBMISSION CHECKLIST

Submission Requirement	Status / Verification
Cover Page & Student Particulars	Formatted & Complete
Certificate of Completion	Ready for Guide Signatures
Declaration & Acknowledgement	Complete
Abstract & Keywords	250–400 Words Complete
Table of Contents & Abbreviations	Updated
All 15 Chapters Complete	100% Comprehensive
Testing Evidence & Defect Resolution	Documented in Chapter 9 & 11
Authentic APA 7th References	Cited & Verified
Source Code Repository	`https://github.com/myusikanani/smart-parking.git`